

Objectives Overview

Define system software and identify the two types of system software

Describe the functions of an operating system

Summarize the startup process on a personal computer

Summarize the features of several stand-alone operating systems

Briefly describe various server operating systems

Summarize the features of several embedded operating systems

Explain the purpose of several utility programs

System Software

- **System software** consists of the programs that control or maintain the operations of the computer and its devices



Operating
systems



Utility
Programs

Operating Systems

- An **operating system (OS)** is a set of programs containing instructions that work together to coordinate all the activities among computer hardware resources

Start and shut
down a
computer

Provide a user
interface

Manage
programs

Manage
memory

Coordinate
tasks

Configure
devices

Establish an
Internet
connection

Monitor
performance

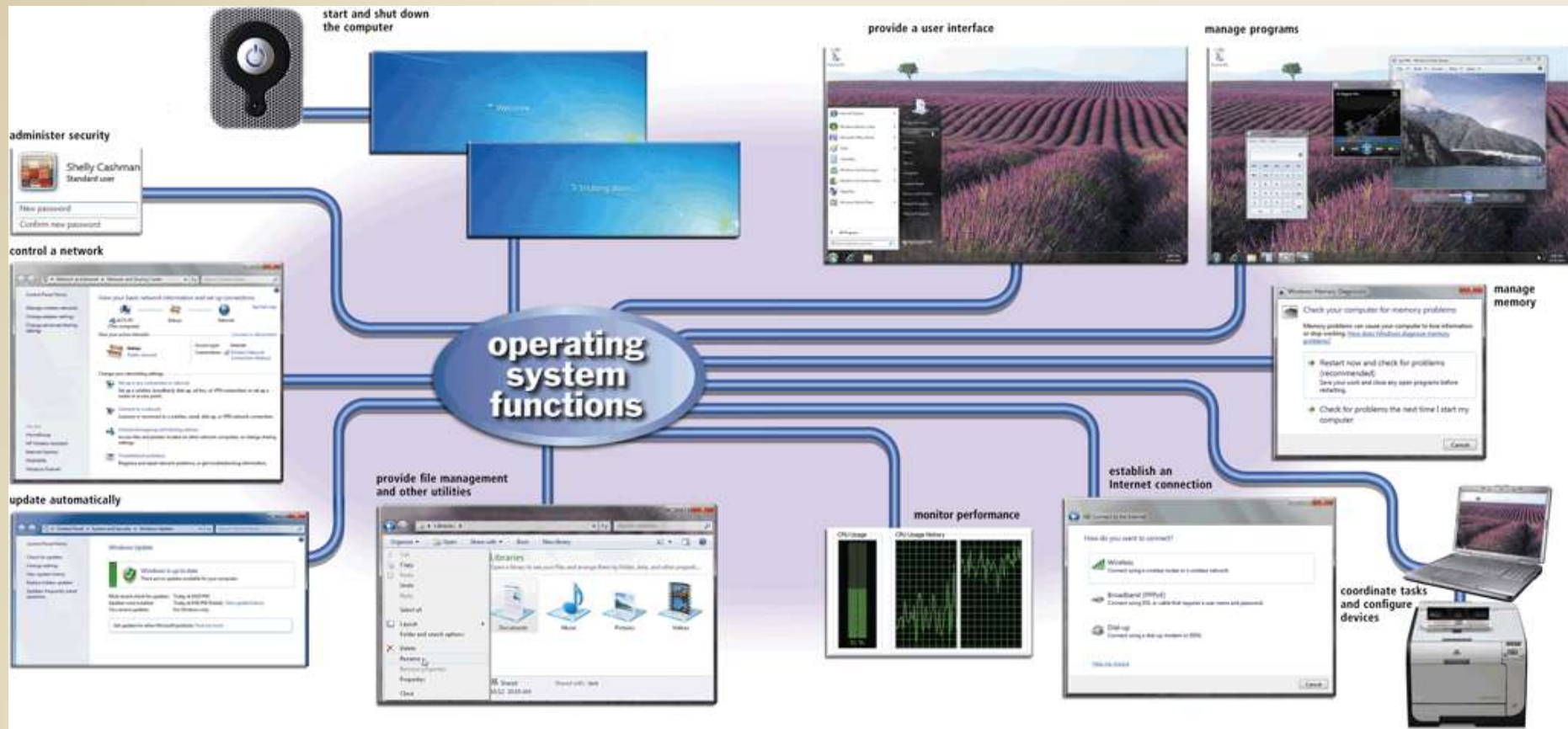
Provide
utilities

Automatically
update

Control a
network

Administer
security

Operating Systems



Operating System Functions

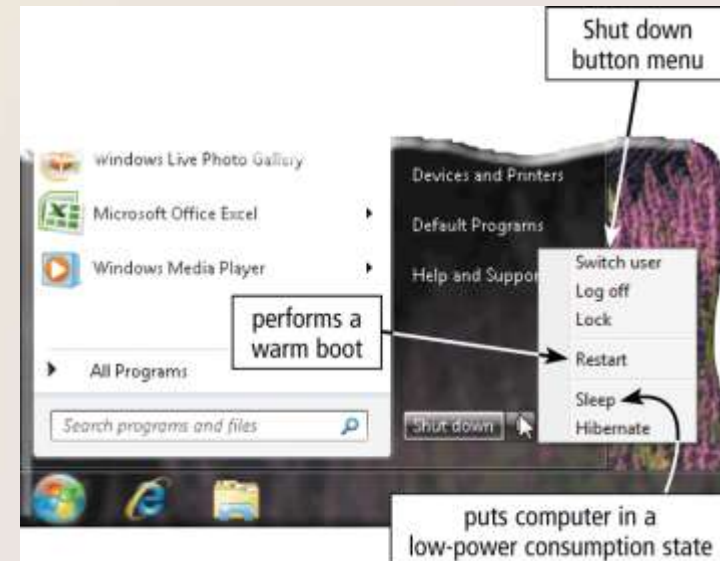
- The process of starting or restarting a computer is called **booting**

Cold boot

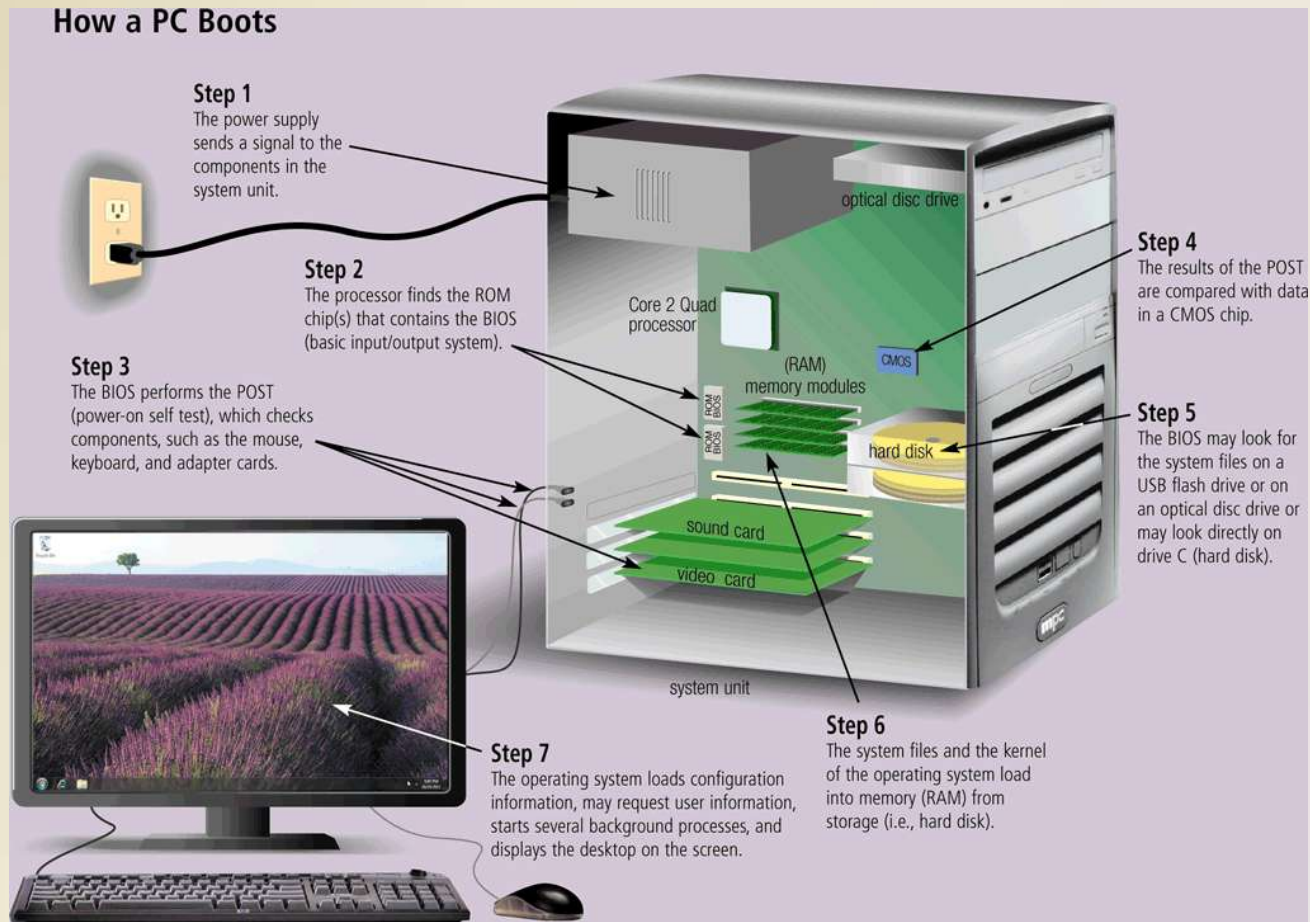
- Turning on a computer that has been powered off completely

Warm boot

- Using the operating system to restart a computer



Operating System Functions



Operating System Functions

A **boot drive** is the drive from which your computer starts

- You can boot from a **boot disk**
- A **recovery disk** contains a few system files that will start the computer

Operating System Functions

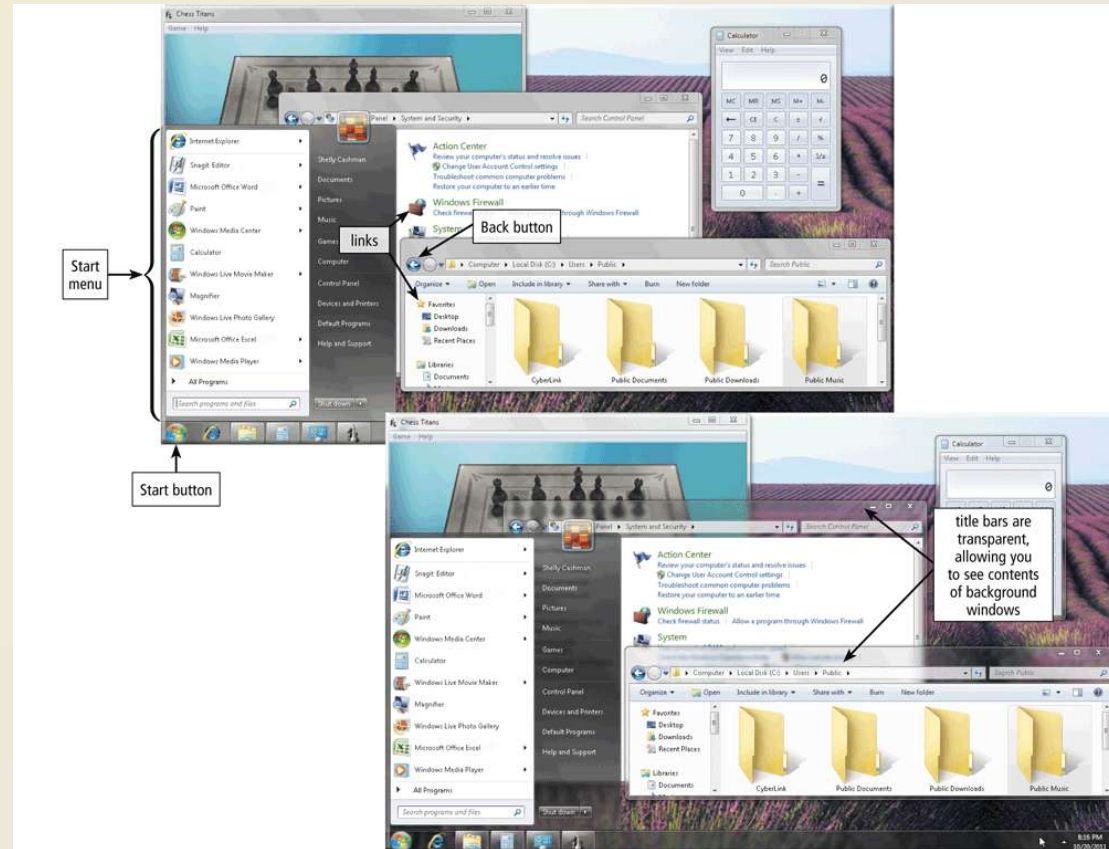
- An operating system includes various shut down options

Sleep mode saves any open documents and programs to RAM, turns off all unneeded functions, and then places the computer in a low-power state

Hibernate saves any open documents and programs to a hard disk before removing power from the computer

Operating System Functions

- A **user interface** controls how you enter data and instructions and how information is displayed on the screen
- With a graphical user interface (GUI), you interact with menus and visual images



Operating System Functions

- With a command-line interface, a user uses the keyboard to enter data and instructions

```
bash-2.05b$ ping -q -c1 en.wikipedia.org
PING rr.chtpa.wikimedia.org (207.142.131.247) 56(84) bytes of data.

--- rr.chtpa.wikimedia.org ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/ndev = 112.076/112.076/112.076/0.000 ms
bash-2.05b$ grep -i /dev/sda /etc/fstab | cut --fields=3
/dev/sda1                /mnt/usbkey
/dev/sda2                /mnt/ipod
bash-2.05b$ date
Wed May 25 11:36:56 PDT
bash-2.05b$ lsmod
Module                  Size  Used by
joydev                  8256  0
ipw2200                 175112  0
ieee80211               44228  1 ipw2200
ieee80211_crypt         4872  2 ipw2200,ieee80211
e1000                   84468  0
bash-2.05b$
```

command entered by user

command prompt

Operating System Functions

- How an operating system handles programs directly affects your productivity

Single user and
multiuser

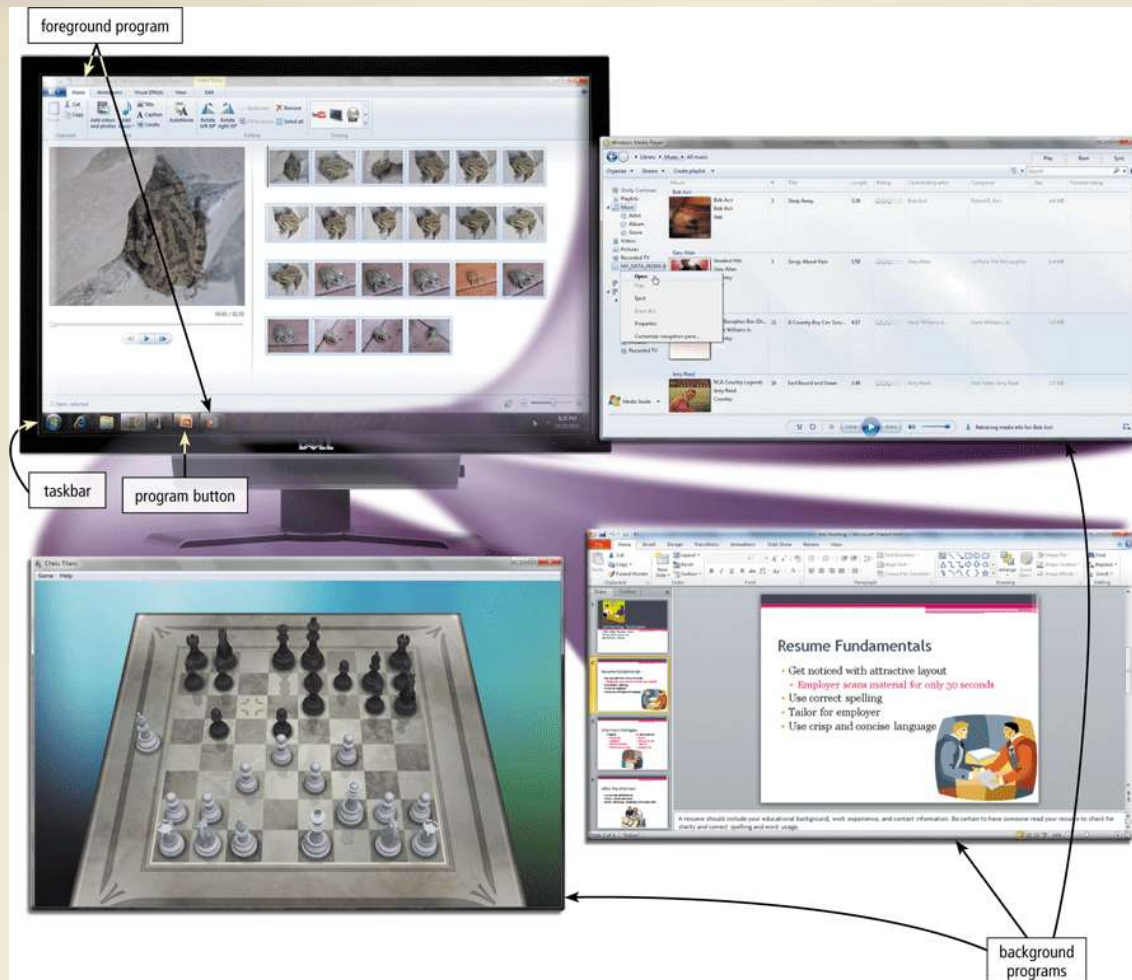
Single tasking
and
multitasking

Foreground and
background

Preemptive
multitasking

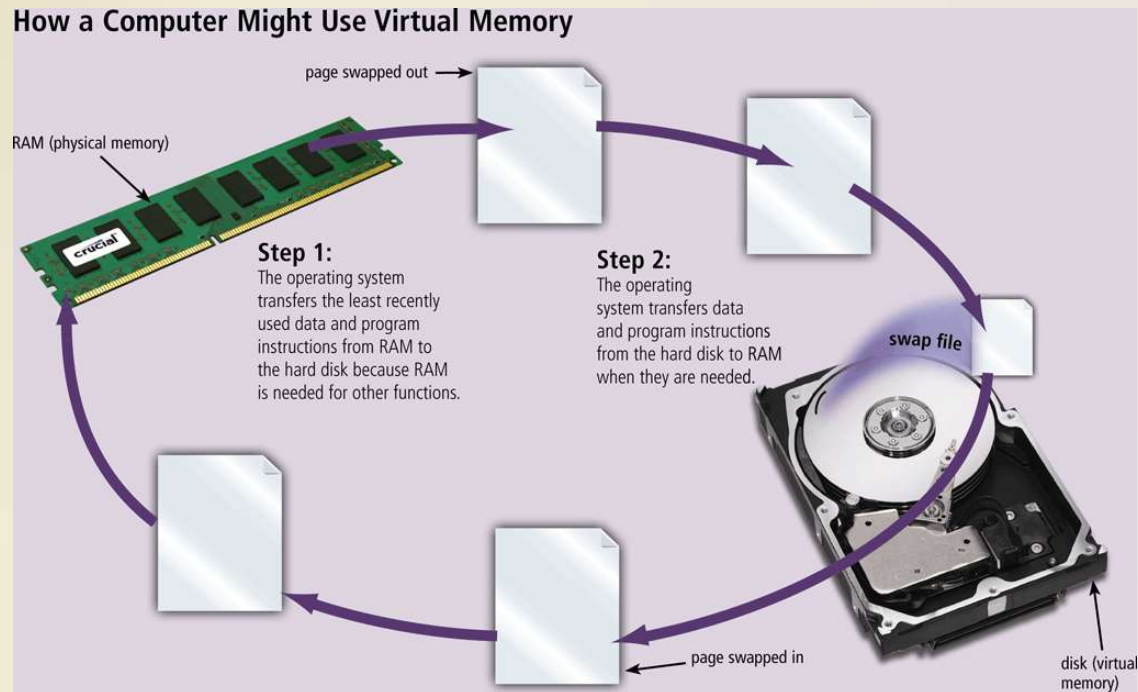
Multiprocessing

Operating System Functions



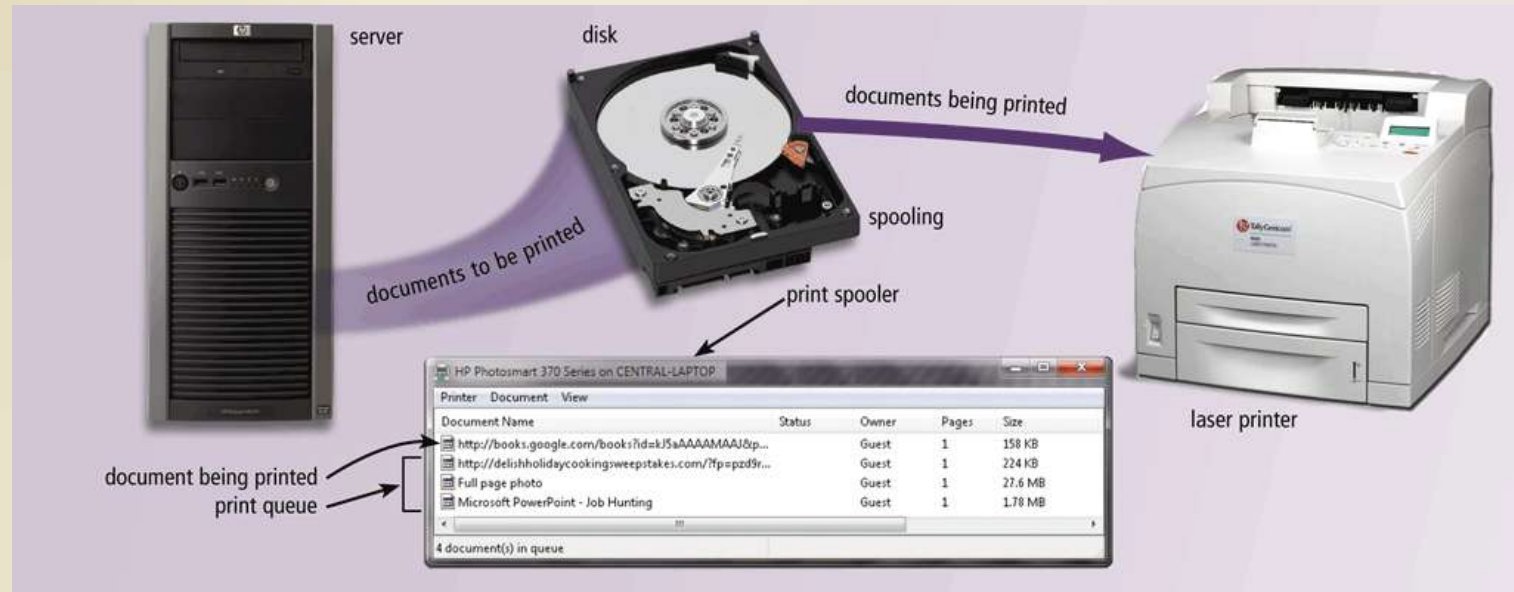
Operating System Functions

- **Memory management** optimizes the use of RAM
- **Virtual memory** is a portion of a storage medium functioning as additional RAM



Operating System Functions

- The operating system determines the order in which tasks are processed



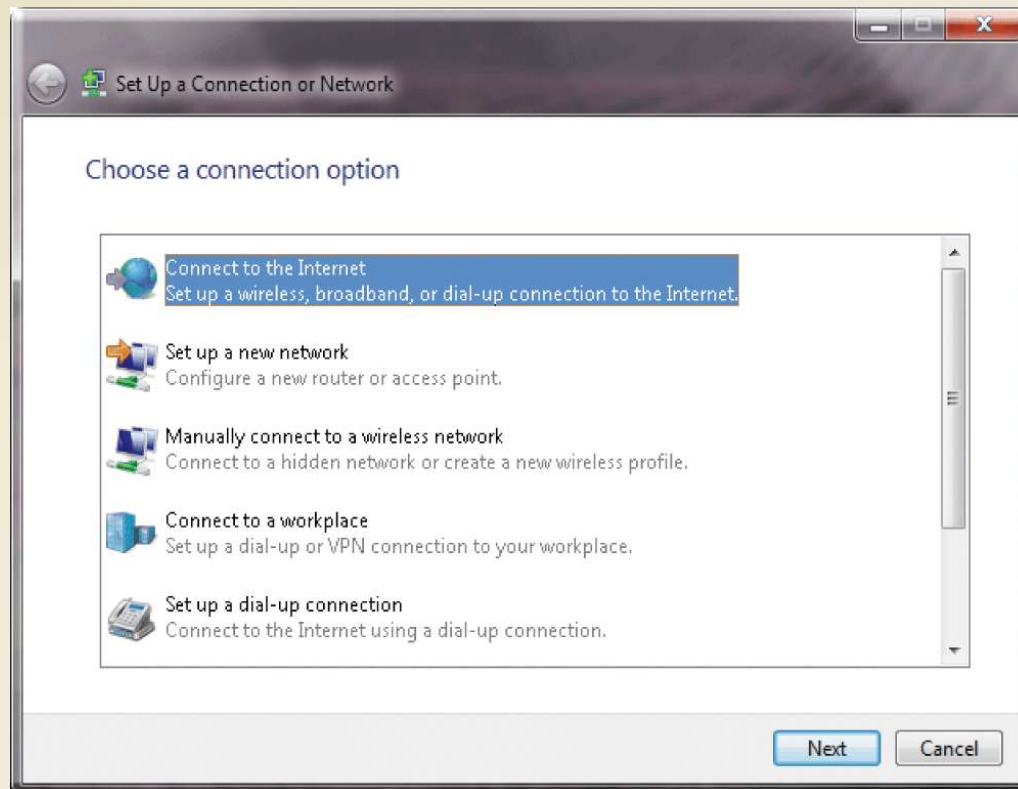
Operating System Functions

A **driver** is a small program that tells the operating system how to communicate with a specific device

Plug and Play automatically configures new devices as you install them

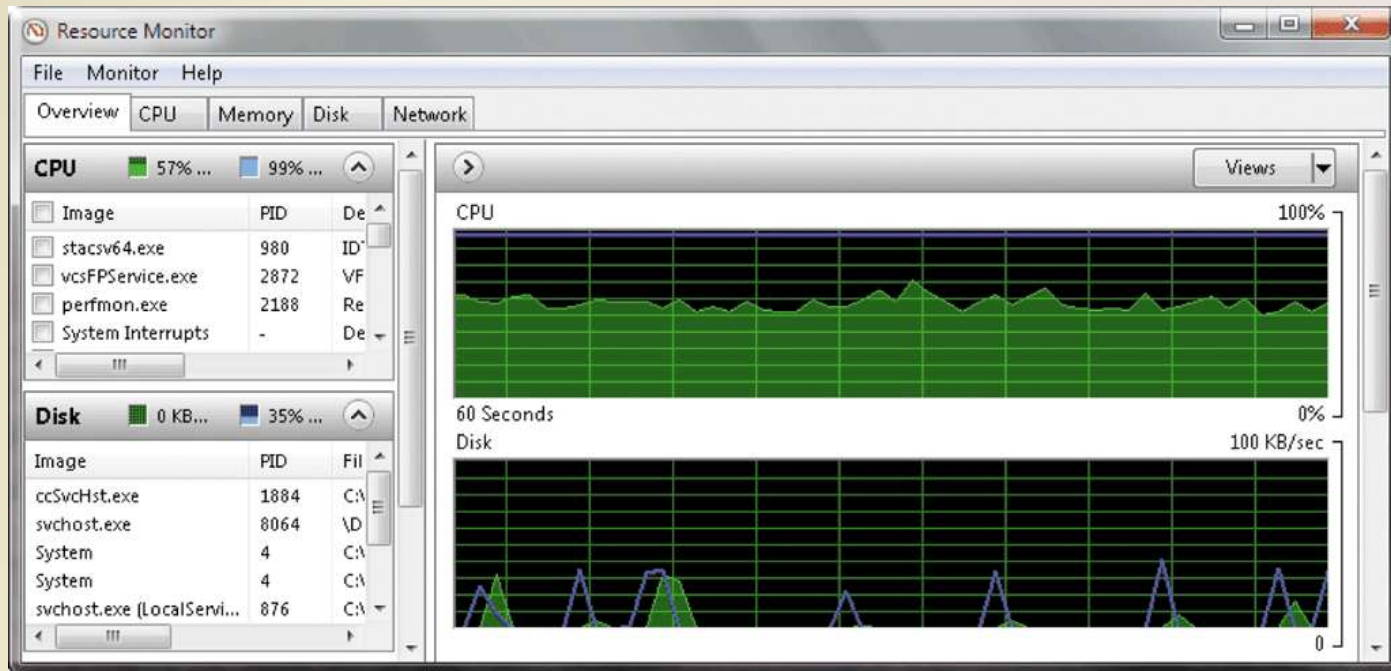
Operating System Functions

- Operating systems typically provide a means to establish Internet connections



Operating System Functions

- A **performance monitor** is a program that assesses and reports information about various computer resources and devices



Operating System Functions

- Operating systems often provide users with the capability of:

Managing files

Searching for
files

Viewing
images

Securing a
computer

Uninstalling
programs

Cleaning up
disks

Defragmenting
disks

Diagnosing
problems

Backing up
files and disks

Setting up
screen savers

Operating System Functions

- **Automatic update** automatically provides updates to the program



Operating System Functions

- Some operating systems are designed to work with a server on a network
- A **server operating system** organizes and coordinates how multiple users access and share resources on a network
- A network administrator uses the server operating system to:
 - Add and remove users, computers, and other devices
 - Install software and administer network security

Operating System Functions

- Each user has a user account
 - A **user name**, or **user ID**, identifies a specific user
 - A **password** is a private combination of characters associated with the user name



Types of Operating Systems

Categories of Operating Systems	
Category	Operating System Name
Stand-alone	• DOS • Early Windows versions (Windows 3.x, Windows 95, Windows NT Workstation, Windows 98, Windows 2000 Professional, Windows Millennium Edition, Windows XP, Windows Vista) • Windows 7 • Mac OS X • UNIX • Linux
Server	• Early Windows Server versions (Windows NT Server, Windows 2000 Server, Windows Server 2003) • Windows Server 2008 • UNIX • Linux • Solaris • NetWare
Embedded	• Windows Embedded CE • Windows Phone 7 • Palm OS • iPhone OS • BlackBerry • Google Android • Embedded Linux • Symbian OS

Stand-Alone Operating Systems

- A **stand-alone operating system** is a complete operating system that works on a desktop computer, notebook computer, or mobile computing device

Windows
7

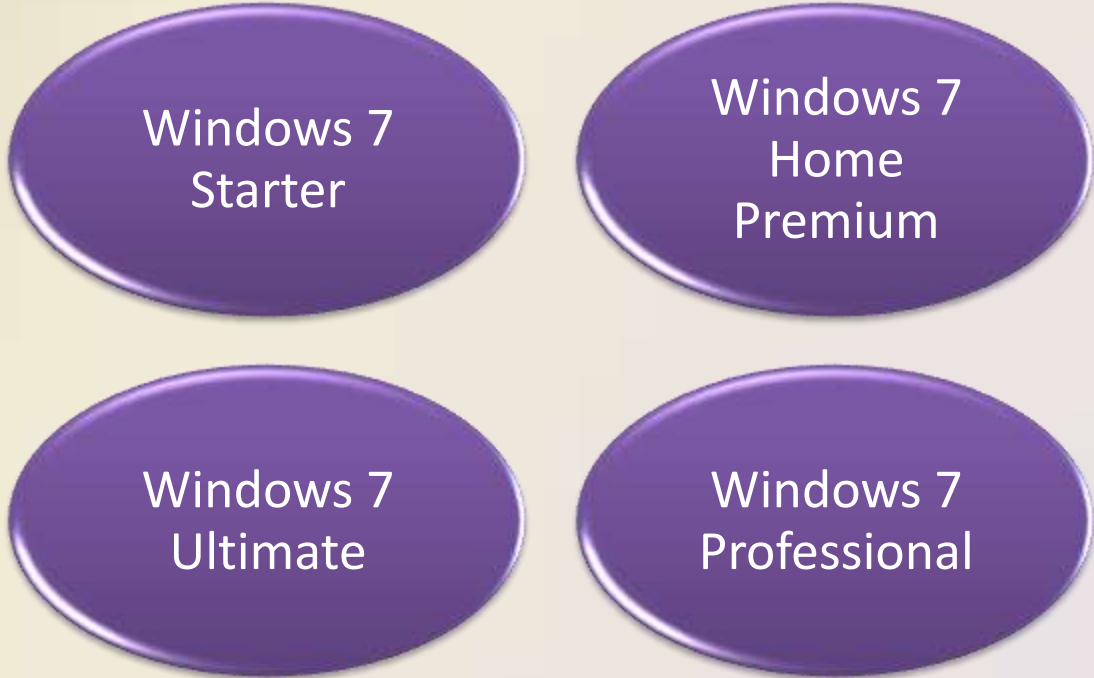
Mac OS X

UNIX

Linux

Stand-Alone Operating Systems

- **Windows 7** is Microsoft's fastest, most efficient operating system to date and is available in multiple editions:



Windows 7
Starter

Windows 7
Home
Premium

Windows 7
Ultimate

Windows 7
Professional

Stand-Alone Operating Systems

- Windows 7 provides programs such as:



Windows
Firewall



Windows DVD
Maker



Windows
Media Player



Desktop
Gadget Gallery

Stand-Alone Operating Systems



Stand-Alone Operating Systems

Additional Windows 7 Features

Reliability and Performance	• Low-powered Sleep state allows you to resume work quickly when you return to the computer • Programs start faster with Windows SuperFetch technology • Automatically detects and fine-tunes performance problems • Built-in hardware diagnostics detect and repair problems automatically • Automatically recovers from failures, including restoring an unbootable computer to a usable state
Security	• User Account Control allows administrators to restrict permissions • Protects users from dangerous Web sites • Improved firewall and <i>Windows Defender</i> protects your computer from external threats • Parental controls allow parents to monitor and control computer usage
Information Management	• Improved and redesigned windows help users locate files by showing thumbnails that preview content • Use Search to locate files based on file name or any other property saved with the file • Coordinate your schedule with others' schedules in <i>Windows Calendar</i> • Use the <i>Snipping Tool</i> to copy screen elements to a file on your computer • Easily share files with other users
Appearance and Navigation	• Easy-to-navigate user interface with translucent windows • Display a preview of windows open in a particular program when you point to that program button on the taskbar • Windows Snap and Aero Shake make it easier to manage the appearance of open windows • Jump Lists make it easier to open recent files quickly in their respective program • <i>Windows Touch</i> supports computers with multi-touch technology
Communications and the Internet	• Enhanced Internet Explorer • Free access to <i>Windows Live Essentials</i>, which includes programs such as <i>Windows Live Messenger</i> for instant messaging, <i>Windows Live Photo Gallery</i> for photo editing and sharing, <i>Windows Live Mail</i> for e-mail, <i>Windows Live Writer</i> for blogging, and <i>Windows Live Movie Maker</i> for video editing and sharing • Consistent and secure wireless network connections • Speech recognition allows you to interact with the computer by voice

Stand-Alone Operating Systems

- The **Macintosh operating system** has set the standard for operating system ease of use
- Latest version is **Mac OS X**



Stand-Alone Operating Systems



UNIX is a multitasking operating system developed in the early 1970s



Linux is an open-source, popular, multitasking UNIX-type operating system

Server Operating Systems

Windows
Server 2008

UNIX

Linux

Solaris

NetWare

Server Operating Systems

- **Windows Server 2008** is an upgrade to Windows Server 2003
 - Part of the **Windows Server 2008 family**
 - Multiple editions
- Most editions include Hyper-V, a virtualization technology
 - Virtualization is the practice of sharing or pooling computing resources

Embedded Operating Systems

- An **embedded operating system** resides on a ROM chip on a mobile device or consumer electronic device

**Windows
Embedded CE**

**Windows
Phone 7**

Palm OS

iPhone OS

BlackBerry

Google
Android

Embedded
Linux

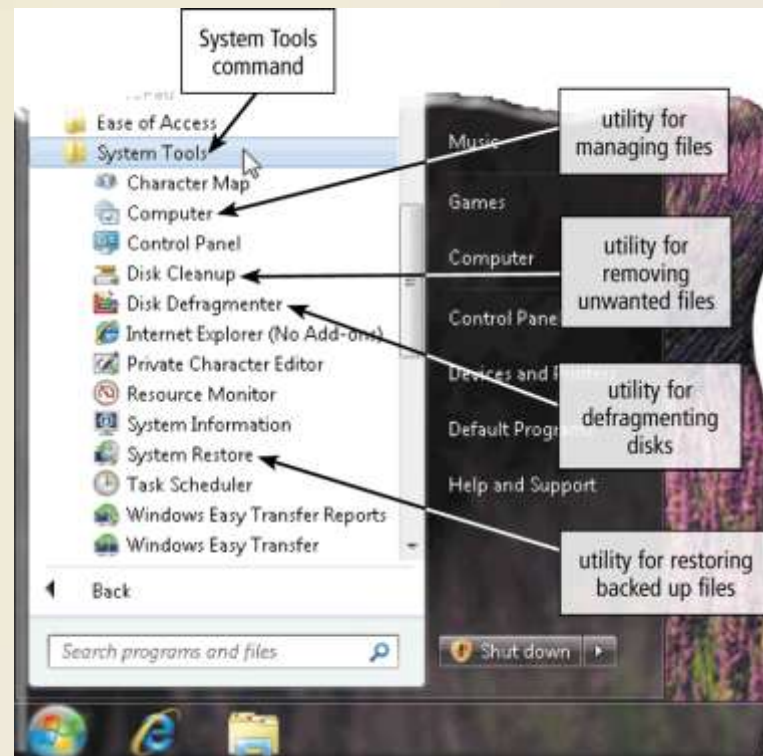
Symbian OS

Embedded Operating Systems



Utility Programs

- A **utility program** is a type of system software that allows a user to perform maintenance-type tasks



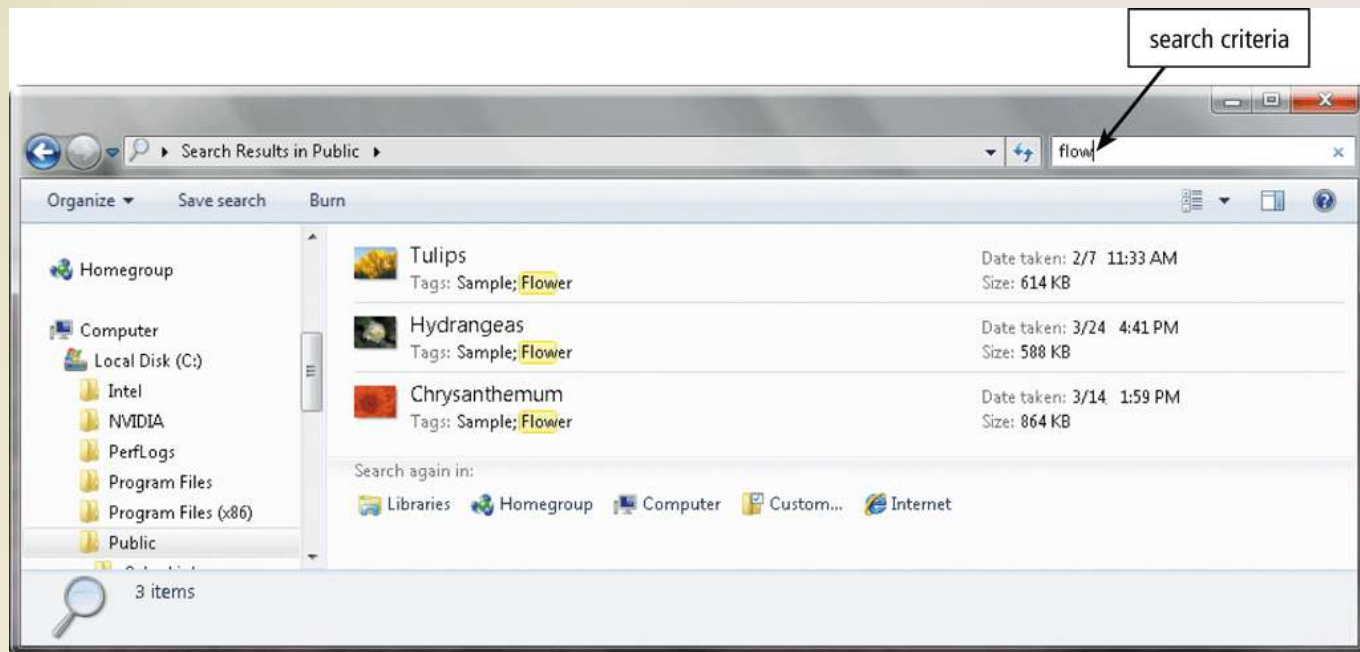
Utility Programs

- A **file manager** is a utility that performs functions related to file management
 - Displaying a list of files
 - Organizing files in folders
 - Copying, renaming, deleting, moving, and sorting files and **folders**
 - Creating **shortcuts**



Utility Programs

- A **search utility** is a program that attempts to locate a file on your computer based on criteria you specify



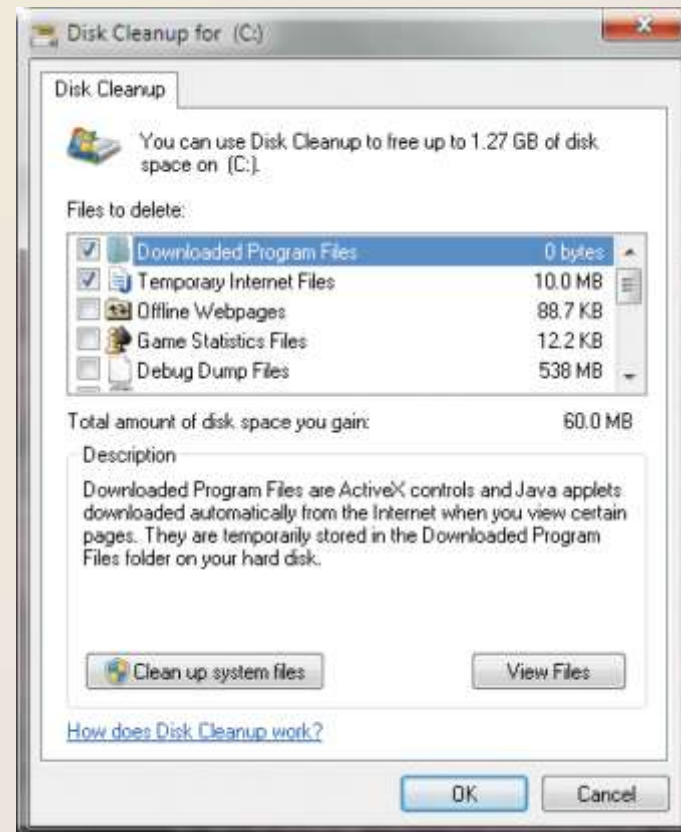
Utility Programs

- An **image viewer** allows users to display, copy, and print the contents of a graphics file
- An **uninstaller** removes a program, as well as any associated entries in the system files



Utility Programs

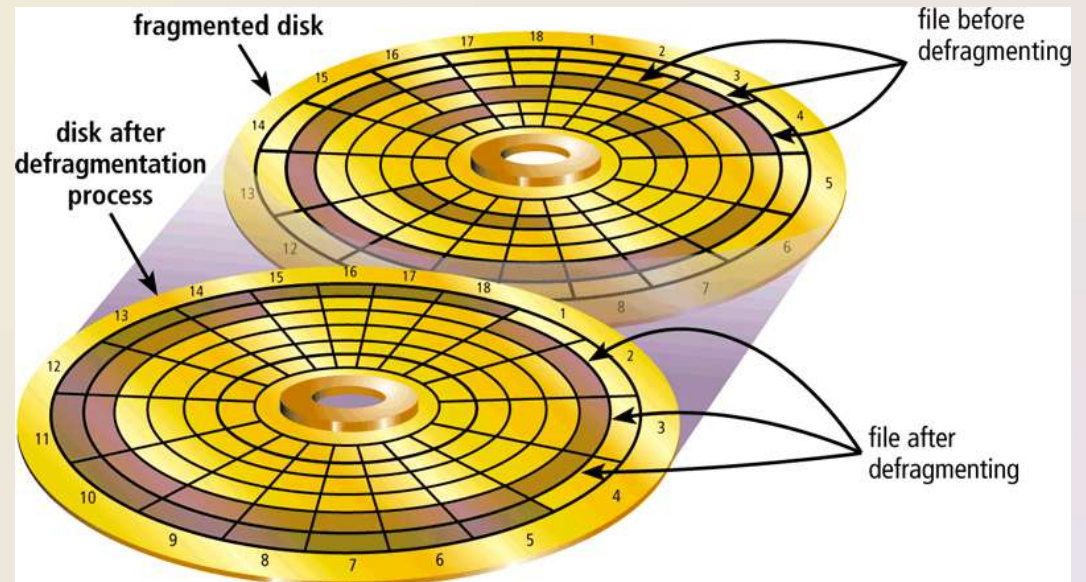
- A disk cleanup utility searches for and removes unnecessary files
 - Downloaded program files
 - Temporary Internet files
 - Deleted files
 - Unused program files



Utility Programs

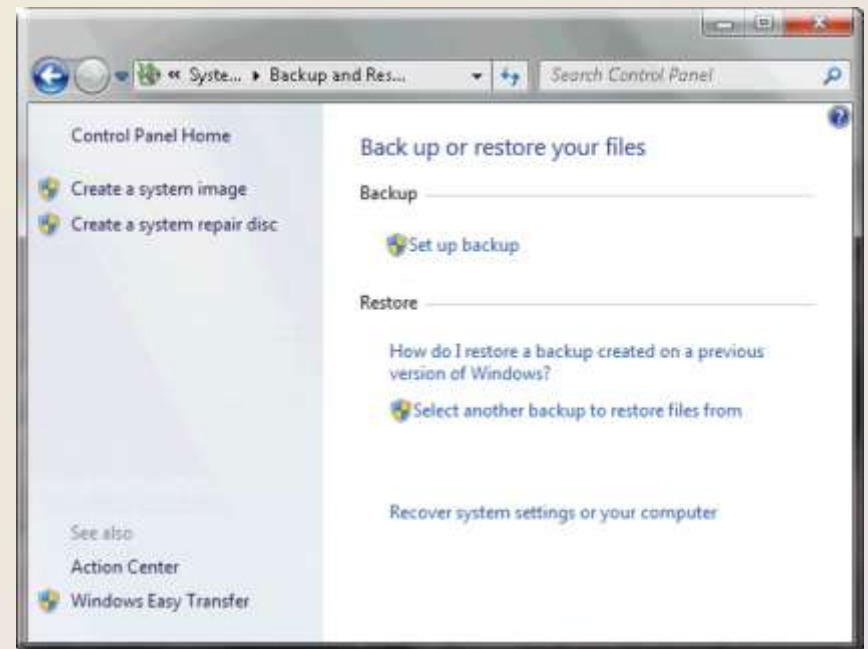
- A **disk defragmenter** reorganizes the files and unused space on a computer's hard disk so that the operating system accesses data more quickly and programs run faster

- **Defragmenting**



Utility Programs

- A **backup utility** allows users to copy files to another storage medium
- A **restore utility** reverses the process and returns backed up files to their original form



Utility Programs

- A **screen saver** causes a display device's screen to show a moving image or blank screen if no activity occurs for a specified time
- A **personal firewall** detects and protects a personal computer from unauthorized intrusions



Utility Programs

A **virus** describes a potentially damaging computer program that affects a computer negatively

A **worm** copies itself repeatedly in memory or over a network

A **Trojan horse** hides within or looks like a legitimate program

An **antivirus program** protects a computer against viruses

Utility Programs

Signs of Virus Infection

- An unusual message or image is displayed on the computer screen
- An unusual sound or music plays randomly
- The available memory is less than what should be available
- A program or file suddenly is missing
- An unknown program or file mysteriously appears
- The size of a file changes without explanation
- A file becomes corrupted
- A program or file does not work properly
- System properties change
- The computer operates much slower than usual



Utility Programs

Spyware is a program placed on a computer without the user's knowledge that secretly collects information about the user

- A **spyware remover** detects and deletes spyware and other similar programs

Adware displays an online advertisement in a banner or pop-up window

- An adware remover is a program that detects and deletes adware

Utility Programs

- Filters are programs that remove or block certain items from being displayed

**Web filtering
software**

**Anti-spam
programs**

**Phishing
filters**

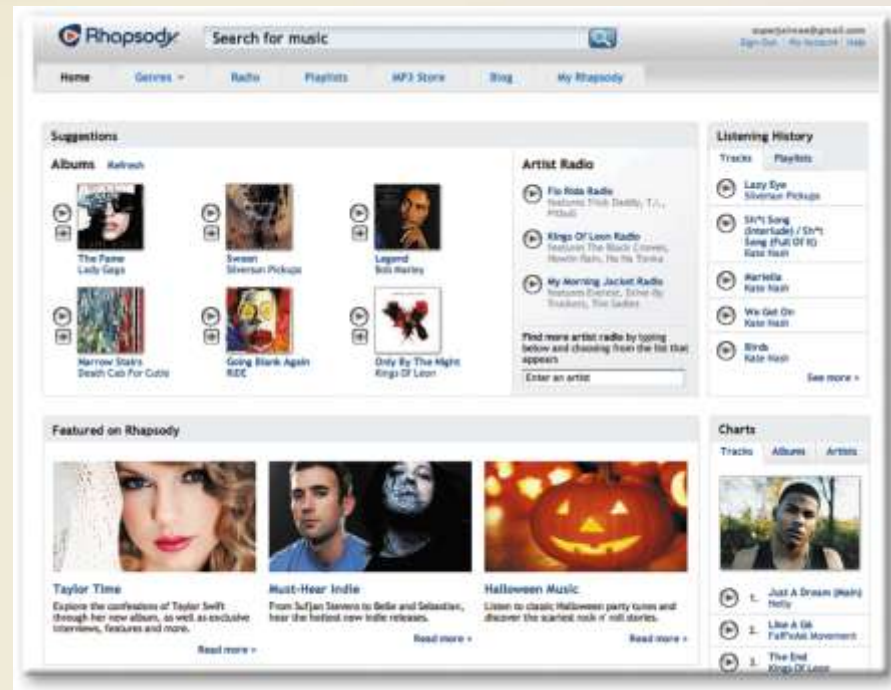
**Pop-up
blockers**

Utility Programs

- A **file compression utility** shrinks the size of a file(s)
 - Compressing files frees up room on the storage media
 - Two types of compression
 - Lossy
 - Lossless
- Compressed files sometimes are called **zipped files**
 - Can be **uncompressed**

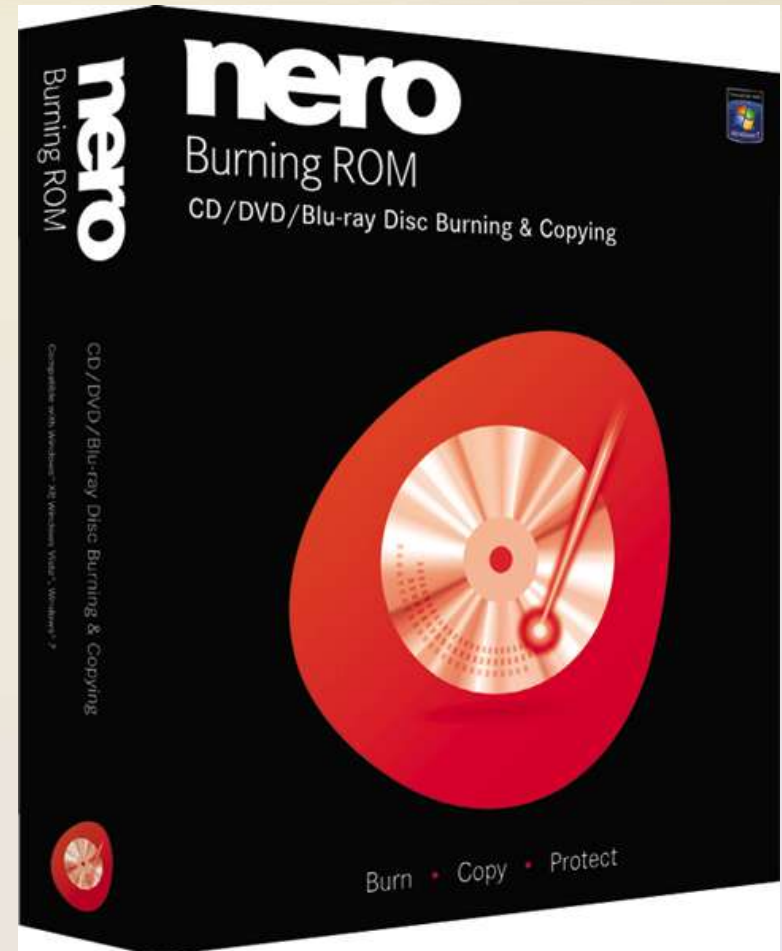
Utility Programs

- A **media player** allows you to view images and animation, listen to audio, and watch video files on your computer

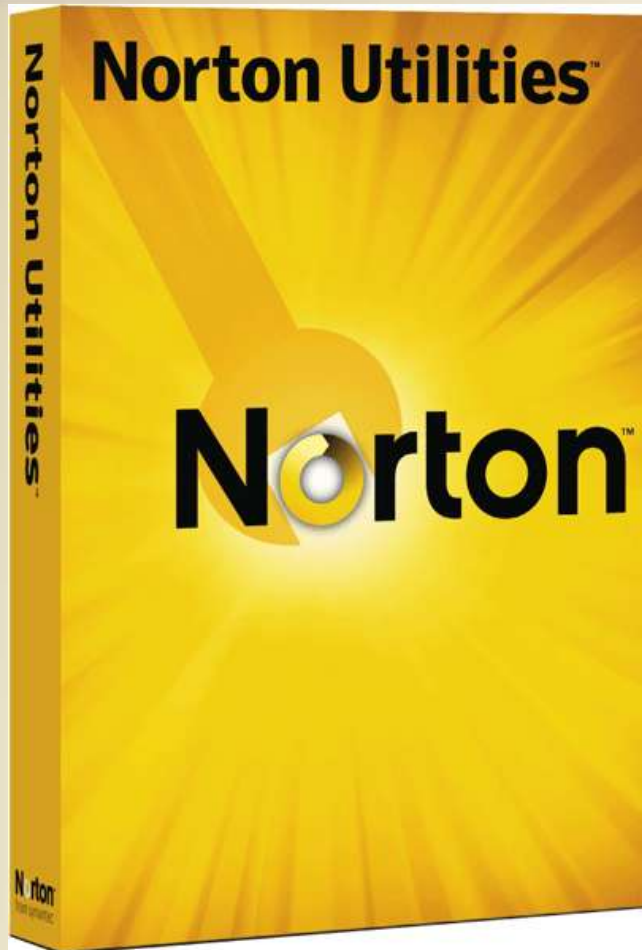


Utility Programs

- **Disc burning software** writes text, graphics, audio, and video files on a recordable or rewritable optical disc



Utility Programs



- A **personal computer maintenance utility** identifies and fixes operating system problems, detects and repairs disk problems, and includes the capability of improving a computer's performance